

(5 pts) 1. List three of the five major mass extinctions that have occurred since the Cambrian. Indicate which event was the largest and which event is the most famous.

a.

b.

c.

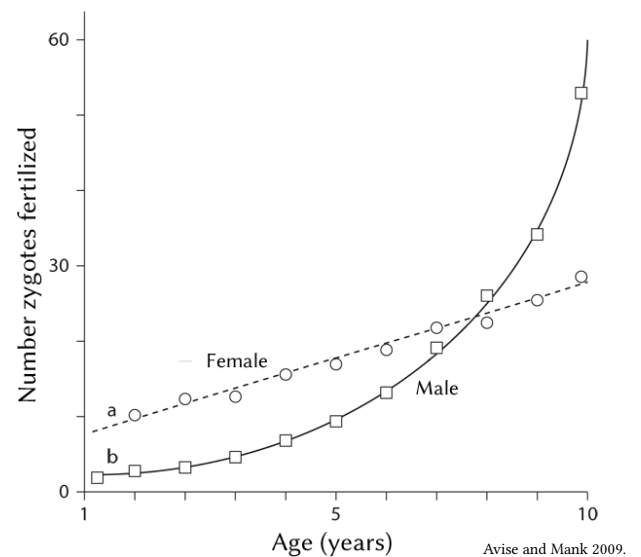
(5 pts) 2. Explain the common, underlying cause of all five mass extinctions that have occurred since the Cambrian.

(5 pts) 3. Describe two adaptive advantages to bipedalism.

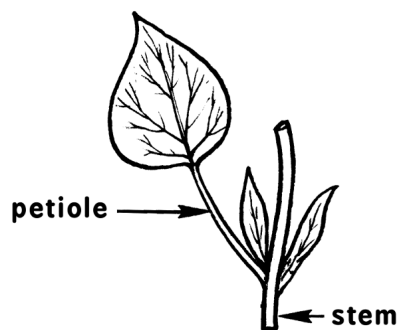
(5 pts) 4. Explain and diagram the cost of sex.

(5 pts) 5. Briefly explain Muller's Ratchet.

(5 pts) 6. Is sequential hermaphroditism likely to evolve in this species? If not, explain why not. If so, then name which type and explain why.



7. Male treehoppers (genus *Enchenopa*) attract females by sending vibrations through plants. One species sends low frequency vibrations through the stem of a plant. The other species sends high frequency vibrations through the petiole. The frequency differences maximize detection by their respective females in their environment (McNett and Cocroft 2008).



(5 pts) (a) Is this intrasexual or intersexual selection? Why?

(10 pts) (b) Did the frequency differences more likely evolved by sensory bias or sensory drive? Explain what is sensory bias or sensory drive (depending on your choice), and then explain how your choice is consistent with the scenario described above.

(5 pts) 8. Explain the Red Queen hypothesis as it relates to the evolution of sexual reproduction as a defense against parasites.

(10 pts) 9. Male satin bowerbirds decorate their bowers with bright blue objects they find around the area. The bower is not a nest. It's a structure built solely to attract females. Females build the nest elsewhere. Assuming the male does *not* help raise his offspring, explain whether the good genes hypothesis or runaway sexual selection best fits this scenario, and why.

(5 pts) 10. Explain the “Replacement Hypothesis” for the migration of early *Homo* into Asia and Europe, and the evidence that supports this hypothesis.

(10 pts) 11. The grasses (Poaceae, includes rice and corn) produce flowers that lack petals and sepals. Given the three classes of MADS genes below, which class or classes most likely had a mutation or change in expression that caused the loss of petals and sepals? Explain why, using the ABC model of floral development. Class A: *Apetala1*. Class B: *Apetala3*. Class c: *Agamous*.

12. Select whether you want **2 points** or **5 points** added to your score for this exam. Before you decide, there is a small catch: if more than 10% of the class selects 5 points, then no one gets the extra points. Only I will see your choice.

A. 2 points. B. 5 points.